Resonance Centrality:

A Composite Metric for Two Distinct Classes of Structural Importance in Animal Social Networks

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Abstract

Betweenness centrality has dominated animal social network analysis for decades, identifying the nodes through which network traffic most efficiently routes. We term these nodes *Grins*. However, a second class of structurally important individual — the node whose local neighborhood is most richly and densely interconnected — is invisible to standard betweenness analysis. We term these nodes *Beescratches*. Here we introduce *resonance centrality*, a composite metric defined as: Resonance = (Rhythm × 0.4) + (Coherence × 0.4) + (Bridging × 0.2), where Rhythm is normalized degree centrality, Coherence is neighborhood density, and Bridging is normalized betweenness centrality. We validate the metric against null-case permutation models across three independent datasets — a bottlenose dolphin social network (Tursiops truncatus, n=62 individuals) and two sociable weaver colony networks (Philetairus socius, n=42 and n=24 individuals) — obtaining effect sizes of 9.73σ to 29.88σ, all at the 100th percentile of 100 scrambled networks. In two of three datasets, the individual identified by resonance centrality as most structurally important is entirely distinct from the individual identified by betweenness centrality, with removal experiments confirming qualitatively different structural roles. In the third, densest dataset, a single individual (DBMREBK, Philetairus socius, Colony Network_658) ranks highest on both metrics simultaneously, constituting the first confirmed instance of a third class we term the *True Grin*. We propose an empirically grounded taxonomy of three network node classes (Grin, Beescratch, True Grin) and discuss implications for the measurement of individual social importance across taxa.

Keywords: *social network analysis, centrality, betweenness centrality, neighborhood density, animal behavior, bottlenose dolphin, sociable weaver*

1. Introduction

Animal social network analysis has transformed our understanding of how individual relationships scale into collective behavior. Since the foundational work of Lusseau and colleagues applying graph theory to wildlife populations, betweenness centrality has emerged as the dominant metric for identifying individuals of disproportionate structural importance within animal social networks (Lusseau et al., 2003; Wey et al., 2008; Croft et al., 2011). Betweenness centrality asks a specific question: through which nodes do the shortest paths between all pairs of other nodes pass? The individual identified by this metric is, in the language we introduce here, the network's routing hub — the node the network depends on for efficient connection of its parts.

This is a meaningful and biologically important question. Betweenness Grins — high-betweenness individuals — are genuinely load-bearing. Remove them and average path lengths increase; the network pays a measurable cost in connection efficiency. Their structural importance has been confirmed across species and social contexts.

But betweenness centrality was designed to answer one question. There is a second question it was not designed to answer: which individual makes their immediate social neighborhood most richly interconnected? Not the node traffic routes through, but the node whose neighbors are most thoroughly connected to each other — whose presence creates a dense, interwoven local structure. We call this second class of individual the Beescratch, named for the bottlenose dolphin (Tursiops truncatus) of that ID in the Doubtful Sound dataset who exemplifies the pattern: highest resonance score, 51st of 62 in betweenness rank.

The Beescratch has always been present in animal social networks. It has not been measured. Standard betweenness analysis produces a ranked list in which the Beescratch may appear nowhere near the top. Their removal does not substantially increase average path lengths. By the conventional metric, they appear unremarkable. But the local social structure they anchor — the density and richness of interconnection in their neighborhood — is the highest in the network. They provide a different kind of structural importance: not routing efficiency, but neighborhood quality.

Here we introduce resonance centrality, a composite metric designed to detect both types of structural importance and to distinguish between them. The metric integrates three components: degree centrality (Rhythm), neighborhood density (Coherence), and betweenness centrality (Bridging), weighted 0.4 / 0.4 / 0.2 respectively. We validate the metric across three independent datasets using null-case permutation methodology, demonstrate the Grin/Beescratch distinction empirically in two datasets, and report the first confirmed instance of a third class — the True Grin — in which a single individual holds the top rank on both metrics simultaneously.

Our contribution is threefold: (1) a validated composite centrality metric that detects what betweenness misses; (2) an empirically grounded taxonomy of three classes of network node distinguished by their structural role; and (3) evidence that these classes are real, consistent across species, and carry distinct implications for network function.

2. Methods

2.1 The Resonance Centrality Formula

Node Resonance Score = (Rhythm × 0.4) + (Coherence × 0.4) + (Bridging × 0.2)

Rhythm (weight: 0.4) is normalized degree centrality: the proportion of all possible connections in the network that node i actually maintains. Rhythm is bounded [0,1] and measures network reach — how many individuals does this node have direct social access to?

Coherence (weight: 0.4) is neighborhood density: of all pairs among node i's direct neighbors, the proportion that are also connected to each other. Formally, Coherence(i) = $2E_i / (k_i(k_i-1))$, where E_i is the number of edges among node i's k_i neighbors. Coherence = 1 when all neighbors are mutually connected; Coherence = 0 when no neighbors are connected to each other. Coherence is equivalent to the clustering coefficient of node i, here reframed as a measure of neighborhood quality rather than network transitivity.

Bridging (weight: 0.2) is normalized betweenness centrality: the proportion of all shortest paths in the network that pass through node i. Bridging measures structural dependence — how much the network relies on this individual to route connection between otherwise separated parts.

The Network Resonance Score is the mean of all individual node scores. All component metrics are normalized to [0,1] prior to combination.

2.2 Weight Justification

The 0.4 / 0.4 / 0.2 weighting was a design choice motivated by two considerations. First, betweenness centrality is a global path metric, requiring a fully resolved network to be stable. In practice, animal social networks are always incompletely sampled; a single unobserved edge can alter shortest-path routing and shift betweenness rankings substantially (Farine et al., 2023). Downweighting Bridging to 0.2 reduces the tool's sensitivity to this known instability.

Second, we treat local social structure as carrying equal theoretical weight to global connectivity. The distinction motivating this work — between routing hubs and quality anchors — requires a metric in which local richness (Coherence) is not subordinate to global connectivity measures. Weighting both Rhythm and Coherence at 0.4 ensures that an individual with high degree and high neighborhood density can achieve a high resonance score even with minimal bridging contribution.

We acknowledge that the specific values 0.4 / 0.4 / 0.2 were not derived from empirical optimization. Alternative weightings may yield superior predictive performance against longitudinal outcome measures such as survival or reproductive success. We present this weighting as a principled starting point and flag it as an open question for future refinement (see Section 4.4).

A second methodological consideration applies specifically to the sociable weaver datasets sourced from the Animal Social Network Repository. These networks were constructed using mark-recapture with bipartite projection from nest chamber co-use records: two individually marked birds share an edge if they were recorded using the same nest chamber within a season, either simultaneously or at different times. This is methodologically distinct from the gambit of the group approach common in cetacean network studies, as connections are chamber-specific and individually recorded rather than inferred from general spatial co-occurrence. However, the 'at different times' component of the edge definition could create associations between birds who never directly co-occurred, potentially inflating local edge density. This limitation is addressed in Section 4.4.

2.3 Null-Case Validation Methodology

To test whether the resonance tool detects genuine biological structure rather than mathematical artifacts of network size or density, we employed a permutation-based null case methodology:

- Compute the Network Resonance Score for the empirical network.
- Generate 100 null networks by randomly permuting edge assignments while preserving total node count and edge count.
- Compute Network Resonance Scores for all 100 null networks.
- Calculate effect size: σ = (empirical score − null mean) / null SD.
- Record percentile rank of empirical network within the null distribution.

A result at the 100th percentile indicates that the empirical network's resonance score exceeded every one of the 100 null networks. For reference, 3σ constitutes conventional scientific significance; 5σ is the particle physics discovery threshold (applied to the Higgs boson confirmation). Our results range from 9.73σ to 29.88σ.

2.4 Datasets

Doubtful Sound Bottlenose Dolphins (Tursiops truncatus)

We analyzed the dolphin social network from Lusseau et al. (2003), consisting of 62 individual dolphins (nodes) and 159 documented association edges in Doubtful Sound, New Zealand. Edges were weighted by association frequency. This network is among the most widely analyzed animal social networks in the literature and provides a well-characterized baseline for method validation.

Sociable Weaver Colony Networks (Philetairus socius)

We analyzed two sociable weaver colony networks obtained from the Animal Social Network Repository (ASNR; Sosa et al., 2021): Network_639 (42 nodes, 152 edges) and Network_658 (24 nodes, 62 edges). Individual birds were identified by color-band codes. Network_658 is substantially denser than Network_639 (edge density 0.226 vs.

0.175), representing a more socially cohesive colony. Both networks were analyzed independently.

2.5 Fragmentation Analysis

To characterize the distinct structural roles of the Grin and Beescratch in each network, we performed single-node removal experiments. We computed average path length (APL) for the full network, then for each network with the highest-betweenness individual removed, and again with the highest-resonance individual removed. The difference in APL change between these two removals operationalizes the distinction between routing efficiency (Grin) and neighborhood quality (Beescratch).

2.6 Code Availability

All analysis code is publicly available at:

https://github.com/barbarajkeiser-MarsLoop/Resonance-Centrality

Key scripts include:

https://github.com/barbarajkeiser-MarsLoop/Resonance-Centrality/blob/main/code_asnr_grin_finder.py

https://github.com/barbarajkeiser-MarsLoop/Resonance-Centrality/blob/main/code_null_case_validation.py

https://github.com/barbarajkeiser-MarsLoop/Resonance-Centrality/blob/main/Dolphin_Resonance.py

The bottlenose dolphin network data (Lusseau et al., 2003) is publicly available:

https://networks.skewed.de/net/dolphins

Sociable weaver colony networks were obtained from the Animal Social Network Repository:

https://github.com/bansallab/asnr

All datasets and code required to reproduce the results reported here are publicly accessible.

Ethics statement: This study used exclusively publicly available datasets. No animals were observed, handled, or disturbed in the course of this research. The bottlenose dolphin data (Lusseau et al., 2003) and sociable weaver colony networks (Sosa et al., 2021; Animal Social Network Repository) were obtained from published sources. No ethical approval was required.

3. Results

3.1 Null-Case Validation: The Resonance Tool Detects Genuine Biological Structure

Across all three datasets, the empirical network resonance score exceeded all 100 null permutations, placing each network at the 100th percentile of its null distribution (Table 1). Effect sizes ranged from 9.73σ (Network_658) to 29.88σ (Network_639), all substantially exceeding conventional thresholds for statistical significance. These results establish that the resonance tool is measuring real biological structure in animal social networks, not mathematical artifacts of network size or edge count.

Table 1. Null-case validation results and node taxonomy across all three datasets.

Dataset	Species	Nodes	Edges	σ (null)	%ile	Betw. Hub	Resonance Anchor	True Grin
Doubtful Sound	Tursiops truncatus	62	159	**17.72σ**	100th	Grin	Beescratch	None
Network_639	Philetairus socius	42	152	**29.88σ**	100th	REMWHEG	YEREEGM	None
Network_658	Philetairus socius	24	62	**9.73σ**	100th	DBMREBK	DBMREBK	**YES ✓**

APL = average path length. σ = effect size in standard deviations above null mean. All datasets at 100th percentile of 100 null permutations.

3.2 The Grin/Beescratch Distinction: Two Datasets Show Complete Role Separation

In both the Doubtful Sound dolphin network and sociable weaver Colony Network_639, the individual identified by resonance centrality as most structurally important is entirely distinct from the individual identified by betweenness centrality.

In the Doubtful Sound network, dolphin “Grin” holds the highest betweenness score (0.2060), while dolphin “Beescratch” holds the highest resonance score (0.4197) with betweenness rank 51 of 62. In Network_639, REMWHEG holds the highest betweenness (0.4277) while YEREEGM holds the highest resonance score (0.4878) at betweenness rank 27 of 42. In both cases, the resonance-dominant and betweenness-dominant individuals are completely non-overlapping, confirming that the two metrics are detecting different structural properties.

The fragmentation analysis confirms that these differences carry functional consequences. In the Doubtful Sound network, removing Grin (the betweenness hub) increases average path length by 0.272 (from 4.112 to 4.384), while removing Beescratch (the resonance anchor) increases path length by only 0.011. The same pattern holds in Network_639: removing the betweenness hub (REMWHEG) increases APL by 0.489; removing the resonance anchor (YEREEGM) increases APL by only 0.021 (Table 2). Grin-type individuals are structurally load-bearing for network routing efficiency. Beescratch-type individuals are not load-bearing by this measure — but their

contribution to local network richness is the highest in the network, a quality that path length does not capture.

Table 2. Fragmentation analysis: average path length change under single-node removal.

Dataset	Full Network APL	Remove Betw. Hub	Remove Res. Anchor	Δ (Hub − Anchor)
Doubtful Sound	4.112	4.384 (+0.272)	4.101 (+0.011)	**0.261**
Network_639	2.445	2.934 (+0.489)	2.466 (+0.021)	**0.468**
Network_658	2.112	2.854 (+0.742)*	2.854 (+0.742)*	**0.000 (same node)**

**In Network_658, DBMREBK is both the betweenness hub and resonance anchor (True Grin); a single removal serves both experimental conditions.*

3.3 The True Grin: First Confirmed Instance of Dual-Role Structural Dominance

In sociable weaver Colony Network_658, the two metrics converge on the same individual. DBMREBK holds both the highest resonance score (0.4948) and the highest betweenness centrality (0.5491) in the network. This is the first confirmed instance of what we term a True Grin: an individual who simultaneously maximizes both routing efficiency and neighborhood quality.

The fragmentation consequence is correspondingly severe. Removing DBMREBK increases average path length from 2.112 to 2.854, an increase of 0.742. This is the largest single-node removal disruption recorded across all three datasets — larger than the betweenness hub removal in either of the other networks. The True Grin's removal costs the network both efficiency and richness simultaneously.

Network_658 is the densest of the three networks analyzed (edge density 0.226, mean degree 5.17 vs. 3.62 for Network_639). We propose that the True Grin may emerge specifically under conditions of high social density within small groups, where the demands of both network functions can converge on a single highly connected individual. This hypothesis requires testing across a broader range of networks.

3.4 Cross-Species Replication

The Grin/Beescratch distinction was identified first in the bottlenose dolphin network (Lusseau et al., 2003) and independently replicated in sociable weaver Colony Network_639, a phylogenetically distant species with a fundamentally different social structure and ecology. That the same qualitative distinction — routing hub distinct from quality anchor — appears in both a cetacean and a passerine bird provides initial cross-species evidence for the generality of the taxonomy. Both datasets show the same directional pattern: Grin removal substantially increases APL; Beescratch removal does not.

4. Discussion

4.1 What Betweenness Centrality Cannot Find

The result we consider most consequential is not the validation effect sizes — though 9.73σ to 29.88σ speaks for itself — but the near-zero overlap between betweenness and resonance rankings in two of three datasets. Betweenness centrality has been the standard tool for identifying structurally important individuals in animal social networks for over two decades. Our results suggest that this tool finds half the picture. The routing hub was already findable. The quality anchor was not.

The Beescratch does not register as important by conventional analysis because their structural contribution is not visible in path-length metrics. Their removal does not increase average path length appreciably. But their neighborhood cohesion score is the highest in the network: among their neighbors, a higher proportion of all possible connections exist than among the neighbors of any other individual. They are anchoring a particularly rich, interwoven local social structure. Whether this translates into measurable outcomes — coalition stability, information propagation speed, resilience following the loss of other individuals — is an open empirical question. But the structure itself is real and consistent across species.

4.2 The True Grin and the Conditions for Dual-Role Dominance

The True Grin — confirmed in a single individual across three datasets — appears to be genuinely rare. We offer a structural hypothesis for its rarity: the routing role and the quality-anchor role may, under most network conditions, be under opposing pressure. A node that maximizes Bridging (betweenness) does so by connecting otherwise separated parts of the network — which implies that its neighborhood consists, at least partly, of individuals who do not themselves connect to each other. This tends to depress Coherence. Conversely, a node whose neighbors are all highly interconnected tends to be embedded in a dense local cluster, which makes global bridging roles less likely.

Network_658 is the smallest and densest network in our dataset. We speculate that True Grins require a specific size-density combination: small enough that one individual can maintain a large proportion of all possible connections, dense enough that those connections are already partly triangulated by other network members. At larger network sizes, the structural tension between the two roles may become irresolvable by any single individual. Testing this hypothesis requires running the tool across the full ASNR database — 1,196 networks across 92 species — to quantify the frequency and network-structural predictors of True Grin emergence.

4.3 Addressing the Sampling Sensitivity Concern

Farine et al. (2023) demonstrated that betweenness centrality is unstable in incompletely sampled animal social networks: because betweenness depends on global shortest-path structure, a single unobserved edge can substantially alter rankings. This is a genuine limitation of betweenness as a primary metric for animal network analysis.

Our design partially addresses this concern by downweighting Bridging to 0.2. The two components carrying 80% of the metric's weight — Rhythm (degree centrality) and Coherence (neighborhood density) — are local metrics. Degree centrality requires only accurate records of an individual's direct connections; neighborhood density requires accurate records of connections among an individual's neighbors. Both are substantially less sensitive to missing edges in distant parts of the network than global path metrics.

However, we acknowledge that this argument applies in degree rather than absolutely. Neighborhood density could theoretically be affected by incomplete sampling of local edges — a formally distinct concern from betweenness sensitivity to global edge missingness. We flag this as an open limitation. A formal sensitivity analysis — introducing synthetic edge noise at varying rates and measuring stability of Coherence rankings — would provide the empirical answer. We identify this as a priority for follow-up work.

4.4 Limitations and Future Directions

Four limitations warrant explicit acknowledgment. First, the 0.4 / 0.4 / 0.2 weighting was arrived at through design reasoning rather than empirical optimization. Validation against longitudinal individual outcome measures (survival, reproductive success, coalition formation) across networks with known resonance scores would provide the strongest evidence for or against the current weights.

Second, the two sociable weaver datasets were constructed using mark-recapture with bipartite projection from nest chamber co-use records — a methodology distinct from the gambit of the group approach used in many cetacean networks. Two individually marked birds share an edge if they used the same chamber within a season, either simultaneously or at different times. The 'at different times' component could create associations between birds who never directly co-occurred, potentially inflating local edge density and affecting Coherence scores. A formal sensitivity analysis, or validation against direct temporal co-occurrence networks such as those from Ferreira et al. (2020, Ecology & Evolution), would provide the strongest test of whether this affects the node-level rankings we report.

Third, we have validated the tool on two species. A cross-taxonomic claim of generality requires analysis across additional species classes. The ASNR repository contains 101 fish networks, 59 reptilian networks, and ant colony networks, all viable targets for the next validation phase.

Fourth, the True Grin rarity claim is supported by one positive instance across three datasets. Quantifying True Grin frequency across the full ASNR and identifying structural predictors of True Grin emergence would substantially strengthen this finding.

All four limitations are tractable. The code is publicly available and applicable to any network in GraphML format. The full ASNR batch analysis and the Ferreira et al. validation are identified as the authors' immediate priorities.

5. Conclusion

We introduce resonance centrality, a composite network metric that detects two classes of structurally important individuals in animal social networks: the Grin (routing hub, identified by high betweenness) and the Beescratch (quality anchor, identified by high resonance with low betweenness). These two roles are empirically separable, functionally distinct as demonstrated by removal experiments, and consistent across two phylogenetically distant species. In a third, denser network, we identify the first confirmed True Grin — an individual maximizing both roles simultaneously — and propose structural conditions under which such dual-role dominance may emerge.

The practical implication is direct: standard betweenness analysis has been finding the Grin all along. The Beescratch — the node whose presence makes every connection around it richer without being structurally load-bearing by path-length metrics — has been present in every network, undetected, for the duration of the field. Resonance centrality finds them.

The tool is publicly available, fully reproducible, and applicable to any network dataset. We invite replication across the full Animal Social Network Repository.

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Code repository: https://github.com/barbarajkeiser-MarsLoop/Resonance-Centrality